

JEE-Main-22-01-2025 (Memory Based)**[EVENING SHIFT]****Physics**

Question: A ball is thrown at 20 m/s at 60° horizontally. The difference in kinetic energy at highest point of trajectory and at the point of release is; (m = mass)

Options:

- (a) 200 m
- (b) 50 m
- (c) 150 m
- (d) 100 m

Answer: (c)

$$K_1 = \frac{1}{2} \times m \times 20 \times 20$$

$$= 200 m$$

$$K_2 = \frac{1}{2} \times m \times 10 \times 10$$

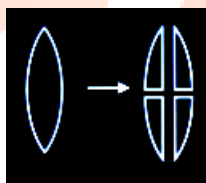
$$= 50 m$$

Question: An equiconvex lens of focal length f , is cut into four parts as shown in the diagram. The focal length of each part is

Options:

- (a) $4f$
- (b) $f/2$
- (c) $2f$
- (d) $f/4$

Answer: (c)



Question: The radius of a tube decreases from $2R$ to R in which ideal liquid is flowing at the same level. Speed at one end is 2 m/s as shown, find speed v at other end

Options:

- (a) 4 m/s
- (b) 1 m/s
- (c) 2 m/s
- (d) 8 m/s

Answer: (c)

Question: Statement 1: Time period of pendulum changes on a planet whose mass is 4 times and radius is 2 times that of earth

Statement 2: Mass of pendulum is same on both planets

Options:

- (a) Both the assertion and reason are true, and the reason is the correct explanation of the assertion
- (b) Both the assertion and reason are true, but the reason is not the correct explanation of the assertion
- (c) The assertion is true, but the reason is false

(d) The assertion is false, but the reason is true

Answer: (d)

Question: The dimensional formula of capacitance is

Options:

(a) $[M^{-1}L^2T^2A^{-3}]$

(b) $[M^{-1}L^{-2}T^4A^3]$

(c) $[M^{-1}L^{-2}T^4A^2]$

(d) $[M^{-1}L^{-2}T^2A^2]$

Answer: (c)

Question: A proton is moving with uniform velocity of 2×10^8 m/s in uniform magnetic and electric fields which are perpendicular to each other. If the electric field is switched off then the proton moves in a circular path of radius 1.6×10^{-5} m. Then magnetic field is:

Options:

(a) 5×10^{-5} T

(b) 1.2×10^5 T

(c) 2.5×10^4 T

(d) 2.5×10^2 T

Answer: (b)

Question: The displacement of a particle moving under the action of a force

$\vec{F} = 2\hat{i} + b\hat{j} + \hat{k}$ is $\vec{d} = \hat{i} + \hat{j} + \hat{k}$. Find the value of b if the work done by the force is zero.

Options:

(a) 0

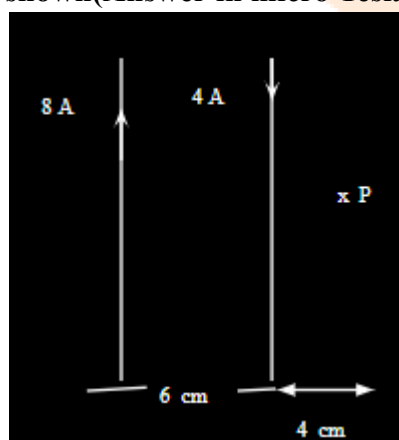
(b) +3

(c) -3

(d) -1

Answer: (c)

Question: Find the Magnetic field at point P due to System of conducting wires as shown (Answer in micro Tesla)



Options:

(a) 1

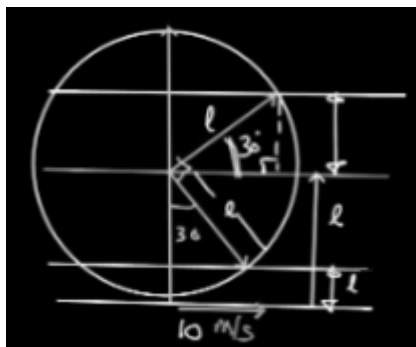
(b) 2

(c) 3

(d) 4

Answer: (b)

Question: A pendulum having a small bob of mass m and length of thread 1m is performing vertical circular as showing in the figure. If its linear velocity of bob at point C is 10 m/s . Find difference between speed of bob at point A and B



Options:

(a) $V_A - V_B = \sqrt{10} \left[\sqrt{8 + \sqrt{3}} + \sqrt{7} \right]$

(b) $V_A - V_B = \sqrt{10} \left[\sqrt{8 - \sqrt{3}} - \sqrt{7} \right]$

(c) $V_A - V_B = \sqrt{10} \left[\sqrt{8 + \sqrt{3}} - \sqrt{7} \right]$

(d) $V_A - V_B = \sqrt{10} \left[\sqrt{7} - \sqrt{8 + \sqrt{3}} \right]$

Answer: (c)

Question: A force is acting along the x axis whose magnitude is changing as per the given equation $F = 8x^2 - 3$. If at an incident work done is 0 then find the variable x .

Options:

(a) $\frac{1}{2\sqrt{2}}$

(b) $\frac{3}{2\sqrt{2}}$

(c) $\sqrt{2}$

(d) 0

Answer: (b)

Question: The point A is situated on the axis of dipole at a distance ' r ' from the dipole with ϵ_0 & V_0 the electric field & electric potential at A. Find electric field & potential at a point B at distance " $2r$ " from dipole on its perpendicular bisector in terms of ϵ_0 & V_0 .

Options:

(a) $\frac{\epsilon_0}{16}, 0 V_0$

- (b) $\frac{\epsilon_0}{16}, 8 V_0$
 (c) $\frac{\epsilon_0}{16}, 3 V_0$
 (d) $\frac{\epsilon_0}{8}, 16 V_0$

Answer: (a)

Question: The maximum percentage error in the measurement of density of a wire is

$$m = (0.60 \pm 0.003) \text{ g}$$

$$r = (0.50 \pm 0.01) \text{ cm}$$

$$l = (10.00 \pm 0.05) \text{ cm}$$

Options:

- (a) 5 %
 (b) 2 %
 (c) 10 %
 (d) 12 %

Answer: (a)

Question: Statement-I : Fringe width of red light is more than fringe width of violet light.

Statement-II : Fringe width is directly proportional to the wavelength of light used.

Choose the correct option.

Options:

- (a) Statement-I is correct and statement-II is incorrect
 (b) Both statement-I and statement-II are correct
 (c) Statement-I is incorrect and statement-II is correct
 (d) Both statement-I and statement-II are incorrect

Answer: (b)

Question: In a photoelectric emission, Radiation of wavelength “ λ ” strikes a plate whose work function is 1 e V and K.E of emitted electron is 4 eV. If the wavelength of electricity radiation becomes half then find the K.E of outcoming electrons.

Options:

- (a) 9 eV
 (b) 10 eV
 (c) 3 eV
 (d) 2 eV

Answer: (a)

Question: A force $\vec{F} = 2\hat{i} + \hat{j} + 2\hat{k}$ acting at a point (1, 1, 1). Find the torque of this force about the origin.

Options:

- (a) $\hat{i} + \hat{j} + \hat{k}$
 (b) $\hat{i} + \hat{j}$
 (c) $\hat{i} - \hat{k}$

(d) $\hat{j} - \hat{k}$

Answer: (c)

Question: For non-vibrating diatomic gas has adiabatic constant of γ_1 & for vibrating diatomic gas has adiabatic constant of γ_2 then

Options:

(a) $\gamma_1 > \gamma_2$

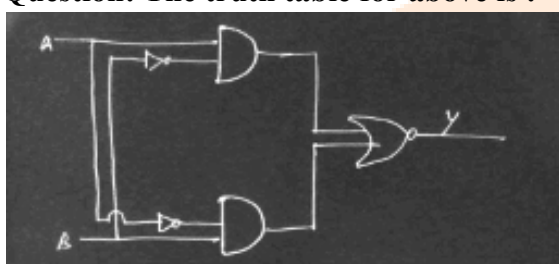
(b) $\gamma_1 < \gamma_2$

(c) $\gamma_1 = \gamma_2$

(d) None of these

Answer: (a)

Question: The truth table for above is :



Options:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(a)

A	B	Y
0	1	1
0	0	0
1	1	0
1	0	1

(b)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

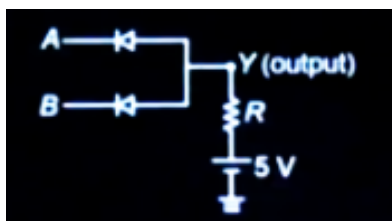
(c)

A	B	Y
0	1	1
0	0	1
1	1	1
1	0	0

(d)

Answer: (a)

Question: Name the logic gate

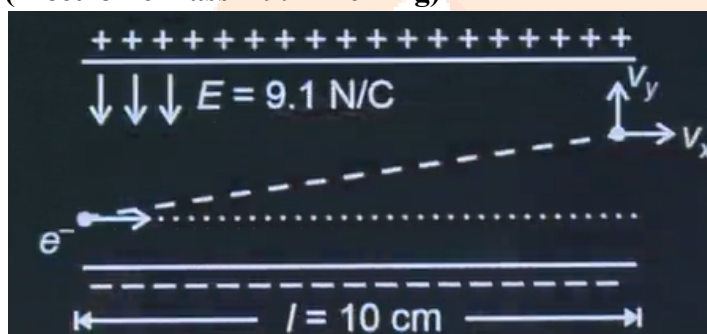


Options:

- (a) OR
- (b) AND
- (c) NOT
- (d) NAND

Answer: (b)

Question: The figure shows an electron entering the space between the plates of a parallel plate capacitor with an initial velocity, $V_x = 10^6$ m/s parallel to the plates. If the length of plates is $l = 10$ cm and the electric field in the region $E = 9.1$ N/C, then the value of v_y when the electron comes out of the plates is (Electronic mass $= 9.1 \times 10^{-31}$ kg)



Options:

- (a) 1.6×10^4 m/s
- (b) 1.6×10^5 m/s
- (c) 1.6×10^7 m/s
- (d) 1.6×10^3 m/s

Answer: (b)

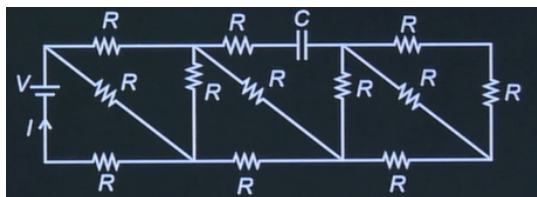
Question: Find the equivalent power of the thin lens combination shown in the figure.

Options:

- (a) $\left(\frac{R_1 + R_2}{R_1 + R_2} \right)$
- (b) $-\left(\frac{R_1 + R_2}{R_1 R_2} \right)$
- (c) $-\left(\frac{R_1 + R_2}{6R_1 R_2} \right)$
- (d) $+\left(\frac{R_1 + R_2}{6R_1 R_2} \right)$

Answer: (c)

Question: In the RC circuit shown, find I .



Options:

- (a) $\frac{V}{5R}$
 (b) $\frac{3R}{8V}$
 (c) $\frac{13R}{3V}$
 (d) $\frac{R}{V}$

Answer: (c)

Question: Solid sphere of mass M , radius R exerts force F on a point mass. Now a concentric spherical mass $M/7$ is removed. What is new force?

Options:

- (a) $\frac{F}{7}$
 (b) $\frac{6}{7}F$
 (c) $\frac{5F}{7}$
 (d) $\frac{3F}{7}$

Answer: (b)

JEE-Main-22-01-2025 (Memory Based)

[EVENING SHIFT]

Chemistry

Question: The density of 3M NaOH is 1.25 g/ml. Molality of solutions is

Options:

- (a) 2.65
- (b) 2.5
- (c) 2.8
- (d) 3

Answer: (a)

Solution:

$$m = \frac{1000 \times M}{1000 \times d - M \times M_w}$$

Question: Arrange according to CFSE.

(i) $[\text{Co}(\text{NH}_3)_4]^{2+}$

(ii) $[\text{Co}(\text{NH}_3)_6]^{3+}$

(iii) $[\text{Co}(\text{NH}_3)_6]^{2+}$

(iv) $[\text{Co}(\text{en})_3]^{3+}$

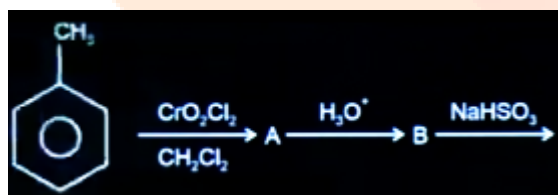
Options:

- (a) (iv) > (ii) > (iii) > (i)
- (b) (iv) > (iii) > (ii) > (i)
- (c) (i) > (iii) > (ii) > (iv)
- (d) (i) > (ii) > (iii) > (iv)

Answer: (a)

Solution:

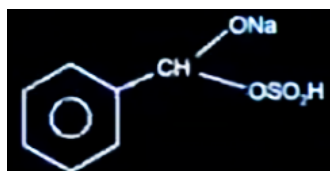
CFSE $\propto$ cation change
 $\propto$ strength of ligand



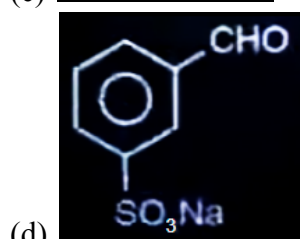
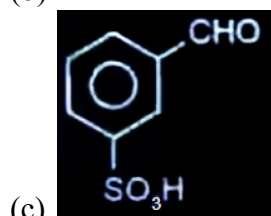
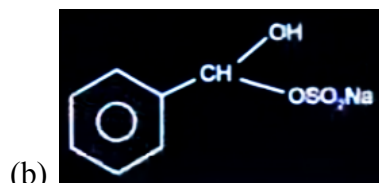
Question:

The product is

Options:

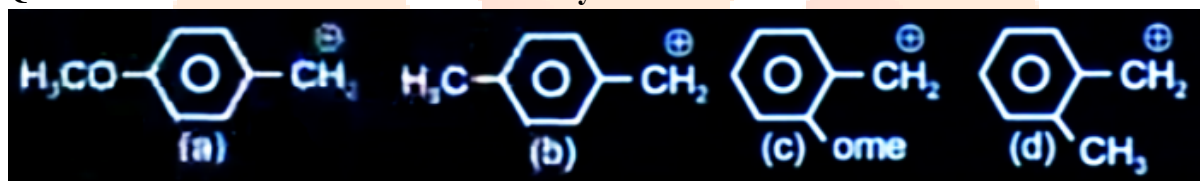


(a)



Answer: (b)

Question: What is correct order of stability of carbocation



Options:

- (a) $a > b > c > d$
- (b) $c > a > d > b$
- (c) $a > c > d > b$
- (d) $c > b > a > d$

Answer: (c)

Question: Compare dipole moment of

(i) NF_3

(ii) CHCl_3

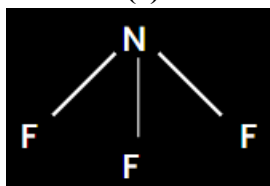
(iii) H_2S

(iv) HBr

Options:

- (a) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (b) $\text{II} > \text{III} > \text{I} > \text{IV}$
- (c) $\text{II} > \text{III} > \text{IV} > \text{I}$
- (d) $\text{III} > \text{I} > \text{IV} > \text{II}$

Answer: (c)



Question: Given below are two statements

S-I: Lassaigue test is used for detection of Nitrogen, phosphorus, sulphur and Halogens.

S-II: Lassaigue extract is made with magnesium metal.

Options:

- (a) Both S-I and S-II are correct
- (b) Both S-I and S-II are incorrect
- (c) S-I is correct but S-II is incorrect
- (d) S-I is incorrect but S-II is correct

Answer: (c)

Question: Which one has two secondary Hydrogen atoms?

Options:

- (a) 2, 2, 4, 4-tetramethylheptane
- (b) 2, 2, 3, 4-tetramethylheptane
- (c) 2, 2, 3, 3-tetramethyloctane
- (d) 3-ethyl-2, 2-dimethylpentane

Answer: (b)

Question: 200 mL of 0.2 M solution of NaOH is mixed with 400 mL of 0.5 M NaOH solution. Molarity of mixture is

Options:

- (a) 0.4
- (b) 0.6
- (c) 4M
- (d) 0.8 M

Answer: (a)

Solution:

$$\frac{200 \times 0.2 + 400 \times 0.5}{600}$$

$$\frac{4 + 20}{60} = \frac{24}{60} = 0.4$$

Question: Which of the following does not show disproportionation reaction

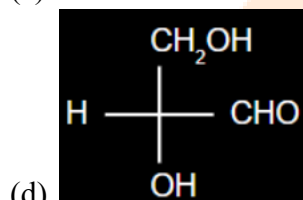
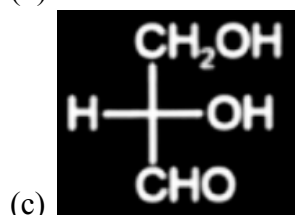
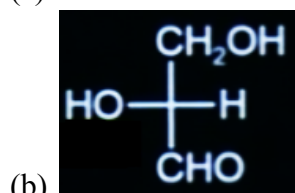
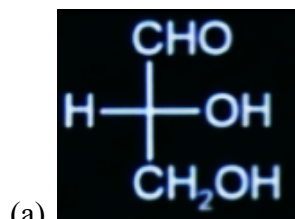
Options:

- (a) ClO_4^-
- (b) ClO_3^-
- (c) ClO_2^-
- (d) ClO^-

Answer: (a)

Question: Correct structure of L-Glyceraldehyde is

Options:



Answer: (b)

Question: Among Group-15 elements, what is the maximum covalency of an element having weakest E-E bond (E = element)

Options:

- (a) 4
- (b) 3
- (c) 5
- (d) 2

Answer: (c)

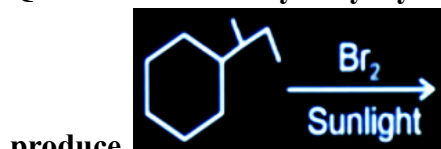
Question: Identify the extensive and intensive property?

Options:

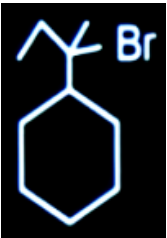
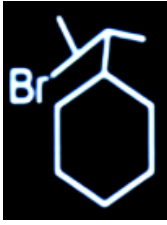
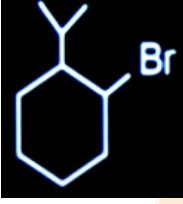
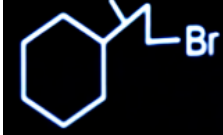
- (a) Mass, volume, conductivity - intensive property
- (b) Mass, temperature, heat, volume - Extensive property
- (c) Mass, volume, internal energy - Extensive property
- (d) Density, temperature, moles internal energy - Extensive property

Answer: (c)

Question: Secondary butyl cyclohexane when reacts with Br₂ in presence of sunlight



Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (a)

Question: What is the relation between K_{sp} and S of $Zr_3(PO_4)_4$

Options:

- (a) $S = \left(\frac{K_{sp}}{6912} \right)^{\frac{1}{7}}$
- (b) $S = \left(\frac{K_{sp}}{144} \right)^{\frac{1}{7}}$
- (c) $S = \frac{K_{sp}}{6912}$
- (d) None

Answer: (a)

Question: Consider the following statements S-1 and S-2 and choose the correct option.

S-1: During corrosion pure metal acts as anode and impure metal acts as cathode.

S-2: Rate of corrosion is more in alkaline medium than in acidic medium

Options:

- (a) Both S-1 and S-2 are correct
- (b) Both S-1 and S-2 are incorrect
- (c) S-1 is correct but S-2 is incorrect
- (d) S-1 is incorrect but S-2 is correct

Answer: (b)

Question: In Ru and Nb, if in Ru, 4d electrons are x and in Nb, 4d electrons are y then find the sum of x and y .

Answer: (7 + 4=11)

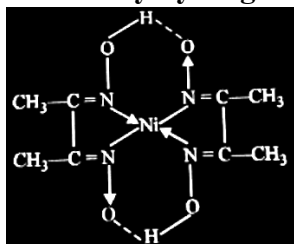
Question: Calculate the radius of first excited state of He^+ ion (in Å)

Answer: (1.058)

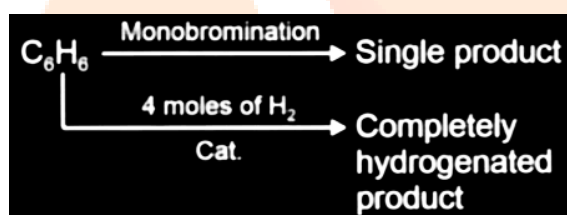
Solution: 0.529×2

Question: $\text{Ni}^{2+} + 2\text{DMG} \rightarrow \text{Complex}$

How many hydrogen bonds are present in a molecule of the complex?



Answer: (2)



Question:
 C_6H_6 .

Find the number of π -electrons in

Answer: (4)



Question: $\text{R} - \text{Br} + \text{Mg} \xrightarrow{\text{dry ether}} \text{A} \xrightarrow{\text{H}_2\text{O}}$
How many $\text{R} - \text{Br}$ can form isopentane?

Answer: (4)

JEE-Main-22-01-2025 (Memory Based)

[EVENING SHIFT]

Maths

Question: 4 boys and 3 girls are to be seated in a row such that all girls seat together and two particular boys B_1 and B_2 are not adjacent to each other. Then the number of ways in which this arrangement can be done.

Options:

- (a) 1002
- (b) 516
- (c) 430
- (d) 432

Answer: (d)

3Girls, 4Boys. $B_1, B_2, B_3, B_4, G_1, G_2, G_3$

$G_3 \text{ Total} - G_3 B_1 B_2$

$$5! \cdot 3! - 4! \cdot 3! \cdot 2!$$

$$= 120 \times 6 - 24 \times 6 \times 2$$

$$= 432$$

Question: If the mean deviation about median for the number 3, 5, 7, $2k$, 12, 16, 21, 24 arranged in ascending order is 6 then the median is

Solution:

$$\text{Median} = \frac{2k+12}{2} = 6 + k$$

$$48 = |k + 3| + |k + 1| + |k - 1| + |6 - k|$$

$$+ |6 - k| + |10 - k| + |15 - k| + |18 - k|$$

$$48 = k + 3 + k + 1 + k - 1 + 6 - k$$

$$+ 6 - k + 10 - k + 15 - k + 18 - k$$

$$48 = -2k + 58$$

$$2k = 10$$

$$k = 5$$

$$\therefore \text{Median} = 11.$$

Question: If $2x^2 + (\cos\theta)x - 1 = 0$, $\theta \in [0, 2\pi]$ has roots α and β . Hence the sum of maximum and minimum value of $\alpha^4 + \beta^4$ is

Options:

- (a) $\frac{25}{16}$
- (b) $\frac{9}{16}$
- (c) $\frac{41}{16}$
- (d) $\frac{1}{17}$

Answer: (a)

$$2x^2 - 1(\cos\theta)x - 1 = 0$$

$$\alpha + \beta = \frac{-\cos\theta}{2}, \alpha\beta = \frac{-1}{2}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \frac{\cos^2\theta}{2} + 1$$

$$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

$$= \left(\frac{\cos^2\theta}{2} + 1\right)^2 - 2\left(\frac{1}{4}\right)$$

$$\text{Min} \rightarrow \cos\theta = 0 \quad \text{Max} \rightarrow \cos\theta = 1$$

$$1 - \frac{1}{2} = \frac{1}{2} \quad \left(\frac{1}{4} + 1\right)^2 - \frac{1}{2}$$

$$\frac{25}{16} - \frac{1}{2} = \frac{17}{16}$$

$$\frac{17}{16} + \frac{1}{2} = \frac{25}{16}$$

Question: Two parallel lines at distance of 5 units, a point p at distance 1 unit from one of the lines. let q be a point on one of the lines and r be the point another one. if the PQR is equilateral triangle, find $(qr)^2$

Solution :

$$\cancel{a^2} - 1 = \cancel{a^2} - r_5 + a^2 - 16 + 2\sqrt{a^2 - 16}\sqrt{a^2 - r_5}$$

$$1600 + a^4 - 80a^2 = 4[a^4 - 41a^2 + 400]$$

$$a^2 = \frac{84}{3}$$

Options:

- Answer: (b)**

$$\begin{aligned}\text{Number of one-to-one functions} &= 256 - 24 \\ &= 232\end{aligned}$$

Options:

- Answer: (d)**

$$(i) 2 \sin^2 \theta = \cos 2\theta$$

$$2 \sin^2 \theta = 1 - 2 \sin^2 \theta$$

$$4 \sin^2 \theta = 1 \rightarrow \sin \theta = \pm \frac{1}{2}$$

$$(ii) 2 \cos^2 \theta = 3 \sin \theta$$

$$2 - 2 \sin^2 \theta = 3 \sin \theta$$

$$\rightarrow 2 \sin^2 \theta + 3 \sin \theta - 2 = 0$$

$$2 \sin \theta (\sin \theta + 2) - 1 (\sin \theta + 2) = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\therefore \sin \theta = \frac{1}{2} \quad \text{or } \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Question: $(1 + y^2) + (x - 2e^{\tan^{-1}y}) \frac{dy}{dx} = 0$

Solution:

$$(1 + y^2) dx = (2e^{\tan^{-1}y} - x^2) dy$$

$$\frac{dx}{dy} = \frac{2e^{\tan^{-1}y}}{1+y^2} - \frac{x}{1+y^2}$$

$$\frac{dx}{dy} + \left(\frac{1}{1+y^2} \right) x = \frac{2e^{\tan^{-1}y}}{1+y^2}$$

$$I.F = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1}y}$$

$$x \cdot e^{\tan^{-1}y} = \int \frac{2e^{\tan^{-1}y} e^{\tan^{-1}y}}{1+y^2} dy$$

$$x \cdot e^{\tan^{-1}y} = \int 2e^{2t} \cdot dt$$

$$x \cdot e^{\tan^{-1}y} = e^{2\tan^{-1}y} + C$$

Question: Perpendicular distance from the point P(-2, 0, 2) to the line

$$\frac{x+1}{2} = \frac{y-1}{-1} = \frac{z+3}{2}$$

Solution:

$$\overline{PM} = (2\lambda + 1, 1 - \lambda, 2\lambda - 5)$$

$$\overline{PM} \cdot \vec{b} = 0$$

$$4\lambda + 12 - 1 + \lambda + 4\lambda - 10 = 0$$

$$7\lambda - 9 = 0$$

$$\lambda = \frac{9}{7}$$

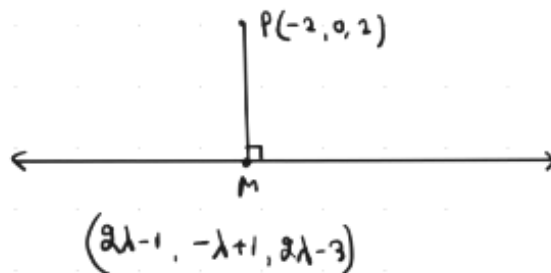
$$M = \left(\frac{18}{7} - 1, 1 - \frac{9}{7}, \frac{18}{7} - 5\right)$$

$$M = \left(\frac{11}{7}, \frac{-2}{7}, \frac{-17}{7}\right)$$

$$PM = \sqrt{\left(-2 - \frac{11}{7}\right)^2 + \left(\frac{2}{7}\right)^2 + \left(2 + \frac{17}{7}\right)^2}$$

$$PM = \sqrt{\left(\frac{-25}{7}\right)^2 + \left(\frac{2}{7}\right)^2 + \left(\frac{31}{7}\right)^2}$$

$$PM = \sqrt{\frac{625+4+961}{49}} = \sqrt{\frac{1590}{7}}$$



Question: Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$.

If $\lambda\vec{a} + 3\vec{b}$ and $2\vec{a} + \lambda\vec{b}$ are perpendicular to each other, then the product of all possible values of λ is _____

Solution :

$$|\vec{a}| = |\vec{b}| = 1$$

$$\text{Now } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \frac{\pi}{3} = \frac{1}{2} \Rightarrow \vec{a} \cdot \vec{b} = \frac{1}{2}$$

$$(\lambda\vec{a} + 3\vec{b}) \cdot (2\vec{a} + \lambda\vec{b}) = 0$$

$$2\lambda + 3\lambda + \lambda^2(\vec{a} \cdot \vec{b}) + 6(\vec{a} \cdot \vec{b}) = 0$$

$$5\lambda + \frac{\lambda^2}{2} + 3 = 0$$

$$\lambda^2 + 10\lambda + 6 = 0$$

$$\text{Produce} = 6$$

Question: The total number of terms in A.P are $2k$. The sum of odd terms is 40 and the sum of even terms is 55 and last term of the A.P exceeds the first term by 27. Then find the value of k .

Solution :

$$T_1, T_2, T_3, \dots, T_{2k}$$

$$T_1 + T_3 + T_5 + \dots + T_{2k-1} = 40$$

$$T_2 + T_4 + T_6 + \dots + T_{2k} = 55$$

$$T_{2k} = T_1 = 27$$

$$\frac{k}{2}[2a + (k-1)2d] = 40$$

$$k[a + (k-1)d] = 40 \dots (1)$$

$$\frac{k}{2}[2(a+d) + (k-1)2d] = 55$$

$$k[a + d + (k-1)d] = 55 \dots (2)$$

$$(2) - (1) \rightarrow dk = 15$$

$$\text{also } a + (2k-1)d - 1 = 27$$

$$(2k-1)d = 27$$

$$2kd - d = 27$$

$$30 - d = 27$$

$$\therefore d = 3$$

$$\text{and } k = 5$$

Question: Consider a function

$$f(x) = \int_0^{x^2} \frac{t^2 - 8t + 15}{e^t} dt.$$

The number of points of

extrema are

Options:

(a) 7

(b) 9

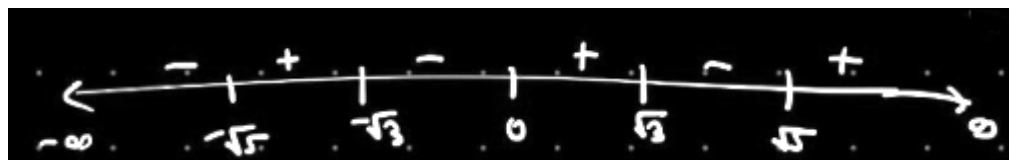
(c) 5

(d) 3

Answer: (c)

$$f'(x) = \frac{x^4 - 8x^2 + 15}{e^{x^2}} 2x$$

$$f'(x) \frac{(x^2-5)(x^2-3)2x}{e^{x^2}} = \frac{(x+\sqrt{5})(x+\sqrt{3})x(x-\sqrt{3})(x-\sqrt{5})}{e^{x^2}}$$



5 points and extrema

Question: Let $A = \{1, 2, 3\}$ then the number of relations on A which consist of ordered pair (1, 2) & (2, 3) and must be reflexive and transitive but not symmetric

Solution:

$$A = \{1, 2, 3\}$$

$$\text{Total relations} = 2^n$$

Must include

$$\{(1, 1)(2, 2)(3, 3)(1, 2)(2, 3)(1, 3)\}$$

$$(2, 1); (3, 2); (3, 1)$$

$$R_1 : \{(1, 1)(2, 2)(3, 3)(1, 2)(2, 3)(1, 3)\}$$

$$R_2 : \{All, (2, 1)\}$$

$$R_3 : \{All(3, 2)\}$$

$$R_4 : \{All, (3, 1)\}$$

$$R_5 : \{All, (2, 1), (3, 2)\}$$

$$R_6 : \{All(2, 1)(3, 1)\}$$

$$R_7 : \{All, (3, 2)(3, 1)\}$$

$$R_8 \{All(2, 1)(3, 2), (3, 1)\}$$

Question: $\lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)^x = \alpha$, find $\frac{\ln \alpha}{1 + \ln \alpha}$

Solution: $\alpha = \lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)^x$ *in form*

$$\log \alpha = \lim_{x \rightarrow \infty} x \log \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)$$

$$\log \alpha = \lim_{x \rightarrow \infty} \left[\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) - 1 \right]$$

if $x \rightarrow 0$

then $\log(1+x) \simeq x$

$$\log \alpha = \lim_{x \rightarrow \infty} \frac{ex}{1-e} \left(\frac{1}{e} - \frac{x}{1+x} \right) - x$$

$$\log \alpha = \lim_{x \rightarrow \infty} \frac{ex}{1-e} \left(\frac{1+x-ex}{e(1+x)} - x \right)$$

$$\log \alpha = \lim_{x \rightarrow \infty} \frac{x}{1+x} \frac{[1+x-ex]}{(1-e)} - x$$

$$\log \alpha = \lim_{x \rightarrow \infty} \frac{x+x^2-ex^2-x(1+x)(1-e)}{(1+x)(1-e)}$$

$$\log \alpha = \lim_{x \rightarrow \infty} \frac{x-x^2-ex^2-(x+x^2)(1-e)}{(1+x)(1-e)}$$

$$\log \alpha = \lim_{x \rightarrow \infty} \frac{x+x^2-ex^2-x-x^2+ex+ex^2}{(1+x)(1-e)}$$

$$\log \alpha = \frac{e}{e} \Rightarrow \frac{\log \alpha}{1+\log \alpha} = \frac{\frac{e}{1-e}}{1+\frac{e}{1-e}}$$

$$= \frac{e}{1-e+e} = e.$$

Question: Let α, β, γ and δ be the coefficient of x^7, x^5, x^3 and x respectively in the expansion of $\left(x + \sqrt{x^2 - 1}\right)^5 + \left(x + \sqrt{x^3 - 1}\right)^5, x > 1$. If u and v

satisfy the equations $\alpha u + \beta v = 18$ $\gamma u + \delta v = 20$ then $u + v$ equals

Options:

(a) 4

(b) 5

(c) 6

(d) 3

Answer: (b)

$$\begin{aligned}
 & \left(x + \sqrt{x^3 - 1} \right)^5 + \left(x + \sqrt{x^3 - 1} \right)^5 \\
 & {}^5C_0 \cdot x^5 + {}^5C_1 \cdot x^4(y) + {}^5C_2 \cdot x^3y^2 + {}^5C_3x^2y^3 + {}^5C_4x^1y^4 + {}^5C_5y^5 \\
 & 2 \left[{}^5C_0x^5 + {}^5C_2x^3 \cdot (x^3 - 1) + {}^5C_4x(x^3 - 1)^2 \right] \\
 & 2 \left[{}^5C_0x^5 + {}^5C_2x^6 - {}^5C_2x^3 + {}^5C_4x(x^6 + 1 - 2x^3) \right] \\
 & 2 \left[{}^5C_0x^5 + {}^5C_2x^6 - {}^5C_2x^3 + {}^5C_4x^7 + {}^5C_4x - 2^5C_4x^2 \right] \\
 & \alpha = 2^5C_4 = 10 \\
 & \beta = 2({}^5C_0) = 20 \\
 & \gamma = 2({}^5C_2) = -20 \\
 & \delta = 2({}^5C_4) = 10 \\
 & 10u + 2v = 18 \\
 & -20u + 10v = 20 \\
 & 14v = 46v = 56 \\
 & v = 4 \\
 & \text{and } 4 = 1
 \end{aligned}$$

Question: Let A and B are two events such that $P(A \cap B) = \frac{1}{10}$ and $P(A/B)$ and $P(B/A)$

are the roots of equation $12x^2 - 7x + 1 = 0$, then $\frac{P(\overline{A} \cup \overline{B})}{P(\overline{A} \cap \overline{B})}$ is equal to

Options:

- (a) 4/9
- (b) 9/4
- (c) 3/2
- (d) 2/3

Answer: (b)